Type IV

Integrals of the form $\int \sqrt{a^2 \pm x^2}\,dx, \int \sqrt{x^2 - a^2}\,dx$

Result 11.3

(1) $\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + c$

(2) $\int \sqrt{x^2 - a^2}\,dx = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\log\left|x + \sqrt{x^2 - a^2}\right| + c$

(3) $\int \sqrt{x^2 + a^2}\,dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\log\left|x + \sqrt{x^2 + a^2}\right| + c$

Proof :

(1) Let $I = \int \sqrt{a^2 - x^2}\,dx$

Take $u = \sqrt{a^2 - x^2}$ then $du = \frac{-2x}{2\sqrt{a^2 - x^2}}\,dx$

$dv = dx \Rightarrow v = x$

Applying Integration by parts, we get

$$\int u\,dv = uv - \int v\,du$$

$$\Rightarrow I = x\sqrt{a^2 - x^2} - \int \frac{-x^2}{\sqrt{a^2 - x^2}}\,dx$$

$$= x\sqrt{a^2 - x^2} - \int \frac{a^2 - x^2 - a^2}{\sqrt{a^2 - x^2}}\,dx$$

$$= x\sqrt{a^2 - x^2} - \int \left(\frac{a^2 - x^2}{\sqrt{a^2 - x^2}} + \frac{(-a^2)}{\sqrt{a^2 - x^2}}\right)dx$$

$$= x\sqrt{a^2 - x^2} - \int \sqrt{a^2 - x^2}\,dx + \int \frac{a^2}{\sqrt{a^2 - x^2}}\,dx$$

$$= x\sqrt{a^2 - x^2} - I + a^2 \int \frac{1}{\sqrt{a^2 - x^2}}\,dx$$

$$2I = x\sqrt{a^2 - x^2} + a^2 \sin^{-1}\left(\frac{x}{a}\right)$$

Therefore, $I = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + c$

Similarly we can prove other two results.

Note 11.3

The above problems can also be solved by substituting $x = a\sin\theta$

Examples 11.41

Evaluate the following :

(i) $\int \sqrt{4 - x^2}\,dx$ (ii) $\int \sqrt{25x^2 - 9}\,dx$ (iii) $\int \sqrt{x^2 + x + 1}\,dx$ (iv) $\int \sqrt{(x-3)(5-x)}\,dx$

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